



Project Proposal
CSE327.7

NGO-Volunteer Coordination Platform

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Project Overview

The NGO & Volunteer Coordination Platform is a web-based application designed to simplify the management of volunteers, events, donations, and team coordination for non-governmental organizations (NGOs). Many NGOs face challenges in organizing volunteers during emergency situations such as floods, cyclones, food distribution drives, blood donation campaigns, or winter clothing distributions. Volunteer information is often stored manually using spreadsheets or paper records, making coordination slow and inefficient.

This project aims to provide a centralized digital platform where NGOs can register volunteers, create events, assign volunteers based on their skills, monitor attendance, track donations, and generate participation certificates. The platform will reduce administrative work while improving communication and resource management.

Volunteers will be able to create personal profiles, list their skills, register for events, and keep track of their volunteer hours. NGOs can create campaigns, recruit volunteers, assign tasks, and monitor participation in real time.

For example, during a flood relief campaign, an NGO can create an event specifying the required number of volunteers and skills such as medical assistance, transportation, logistics, or cooking. Registered volunteers who possess the required skills can apply for the event. Organizers can approve volunteers, divide them into teams, mark attendance on the event day, and automatically generate certificates once the event is completed.

The system will also include donation tracking to help NGOs maintain records of incoming donations and their allocation. This improves transparency and accountability.

To encourage long-term participation, the platform will maintain volunteer statistics, total volunteer hours, and a leaderboard that recognizes active contributors. These features motivate volunteers while helping NGOs identify experienced individuals for future events.

Overall, the NGO & Volunteer Coordination Platform seeks to modernize NGO operations through digital transformation, making volunteer management faster, more organized, transparent, and scalable.

End Users

The primary end users of the system are:

NGO Administrators

- Register and manage their organization.
- Create and manage volunteer events.
- Assign volunteers to teams.
- Track attendance and participation.
- Manage donations.
- Generate reports and certificates.

Volunteers

- Create volunteer profiles.
- Register for events.
- Display skills and interests.
- View assigned teams.
- Track volunteer hours.
- Download participation certificates.

Event Coordinators

- Monitor event progress.
- Mark attendance.
- Manage team assignments.
- Communicate with volunteers.

Donors

- View NGO campaigns.
- Make donations.
- Track donation acknowledgements.

Functional Requirements (Specifications)

User Management

- User registration and login.
- Secure authentication.
- Password reset.
- User profile management.

Volunteer Management

- Volunteer registration.
- Update volunteer information.
- Store volunteer skills.
- Search volunteers by skills.
- View volunteer history.

Event Management

- Create volunteer events.
- Edit and delete events.
- Event scheduling.
- Event registration.
- View event participants.

Skill Matching

- Store volunteer skills.
- Recommend suitable volunteers for events.
- Filter volunteers based on required skills.

Attendance Tracking

- Record attendance.
- Maintain attendance history.
- Generate attendance reports.

Team Assignment

- Divide volunteers into teams.
- Assign team leaders.
- Update team assignments.

Certificate Generation

- Automatically generate certificates after event completion.
- Allow volunteers to download certificates.

Donation Tracking

- Record donations.
- Track donor information.
- Generate donation reports.

Dashboard

- Display upcoming events.
- Display volunteer statistics.
- Display donation summaries.
- Show completed events.

Volunteer Hours

- Automatically calculate volunteer hours.
- Maintain total lifetime volunteer hours.
- Display progress on volunteer profile

Non-Functional Requirements (Specifications)

Performance

- Support hundreds of simultaneous users.
- Load pages within 2–3 seconds under normal conditions.
- Efficient database queries for quick search and retrieval.

Security

- Secure login authentication.
- Password encryption using hashing.
- Role-based access control (Admin, Coordinator, Volunteer).
- Protection against SQL Injection and Cross-Site Scripting (XSS).

Reliability

- High system availability.
- Regular database backups.
- Recovery from unexpected failures.

Scalability

- Support multiple NGOs.
- Handle increasing numbers of volunteers and events.
- Allow future expansion with additional modules.

Usability

- Simple and intuitive user interface.

- Mobile-friendly responsive design.

Maintainability

- Modular software architecture.
- Well-documented code.
- Easy addition of new features.

Availability

- Accessible 24/7 through a web browser.
- Compatible with modern browsers such as Chrome, Firefox, Edge, and Safari.

Compatibility

- Support Windows, Linux, macOS, Android, and iOS devices.
- Responsive layout for desktops, tablets, and smartphones.

Data Integrity

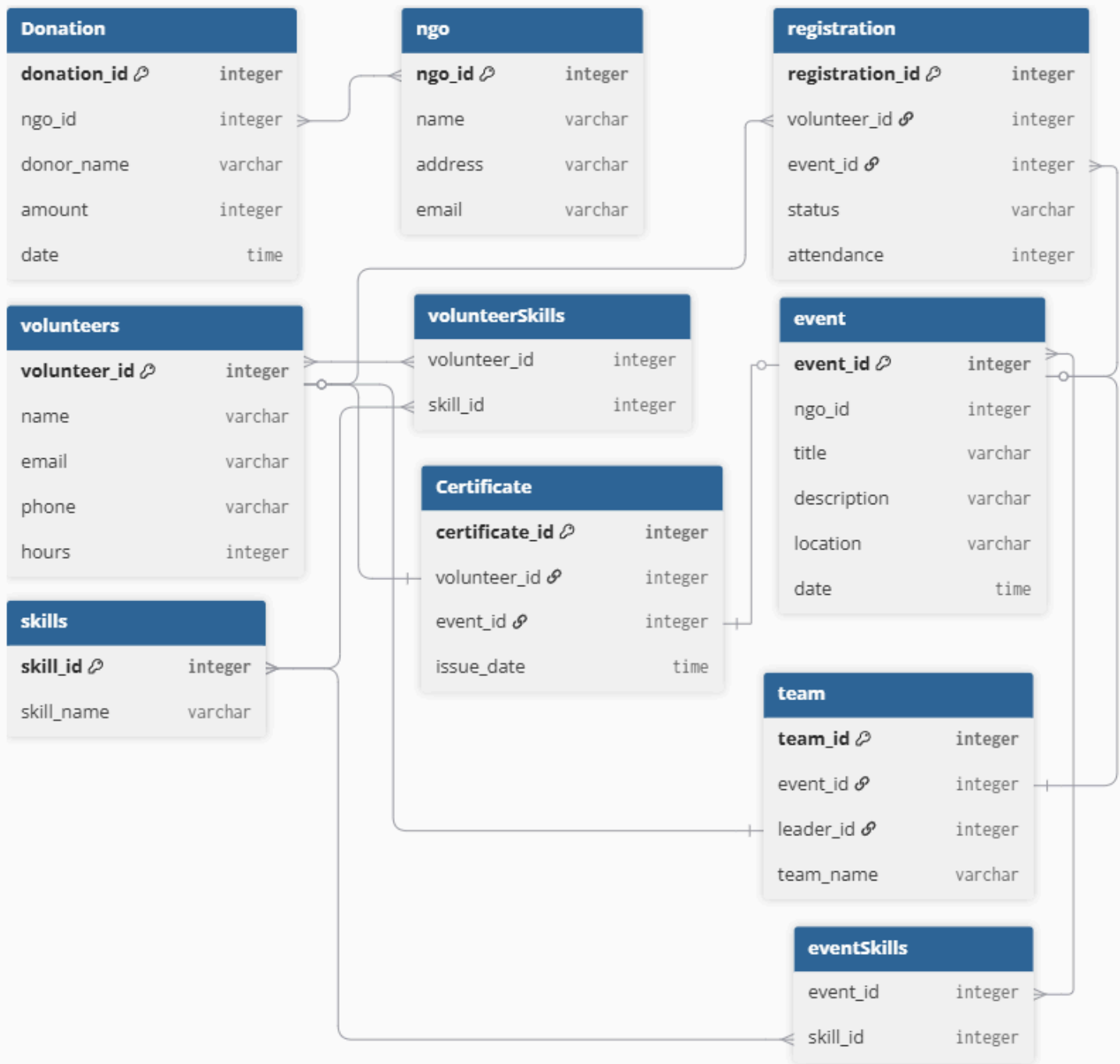
- Ensure accurate storage of volunteer, event, and donation data.
- Prevent duplicate registrations.
- Validate user input before database storage.

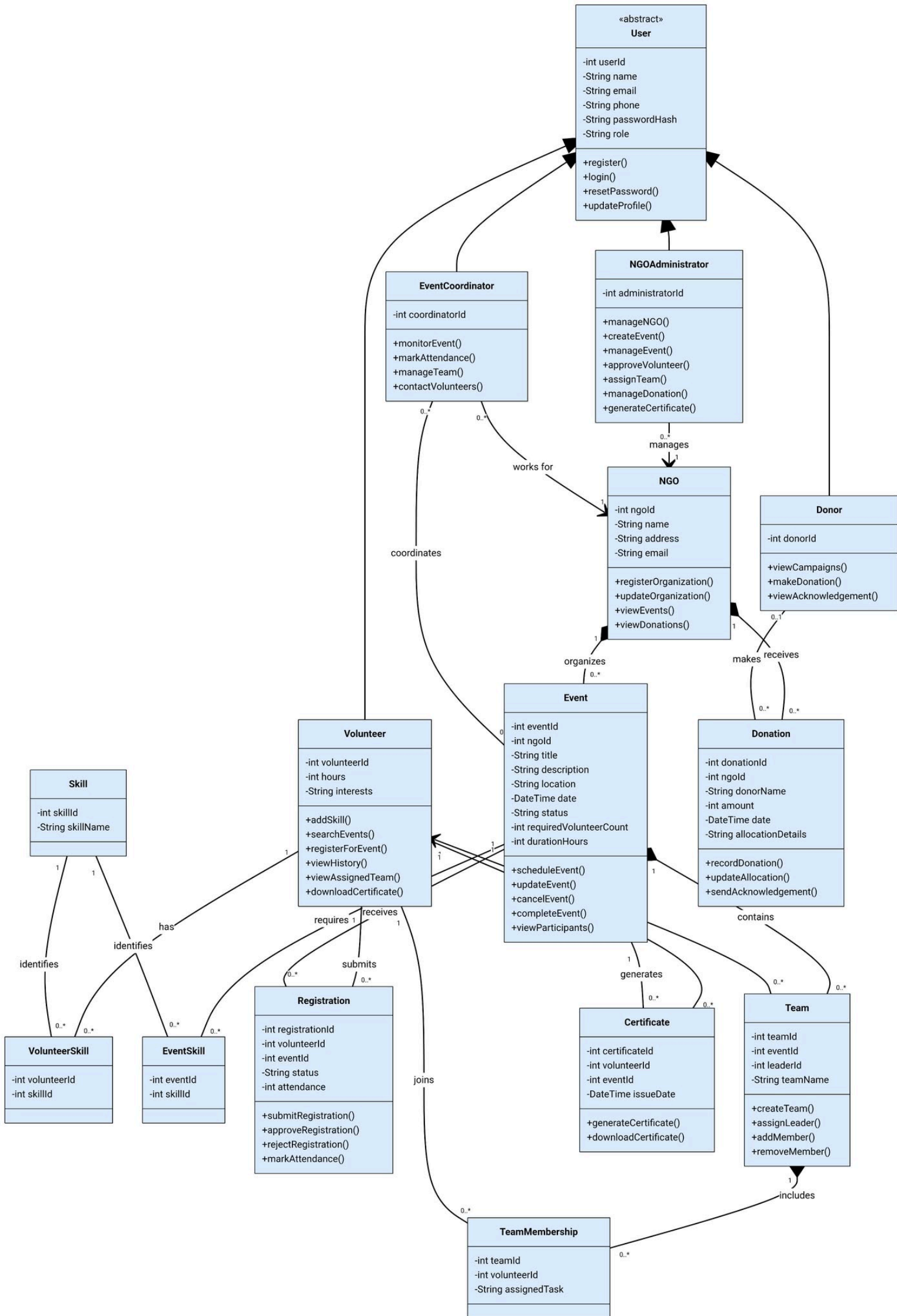
Backup and Recovery

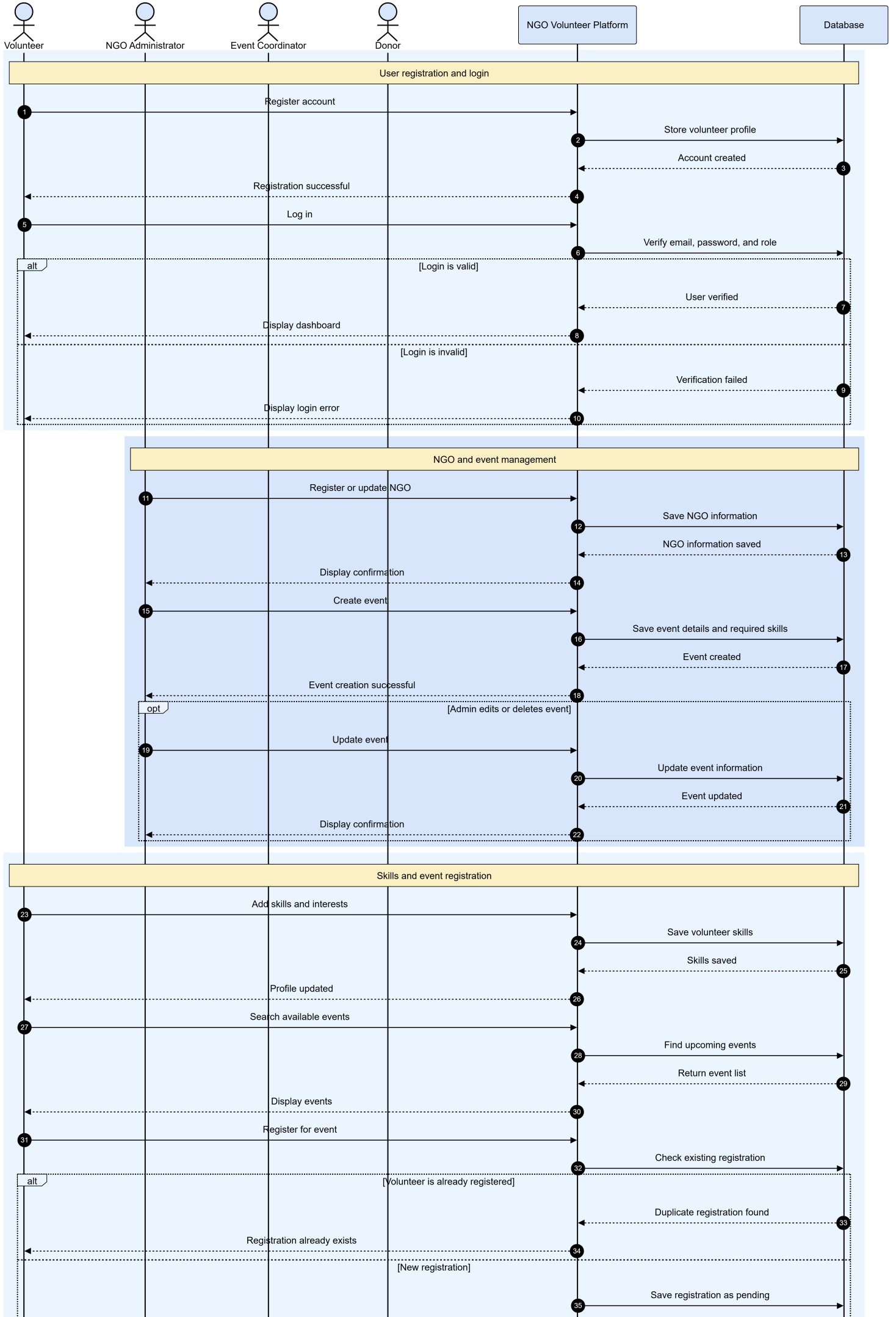
- Scheduled database backups.
- Ability to restore lost data after system failure

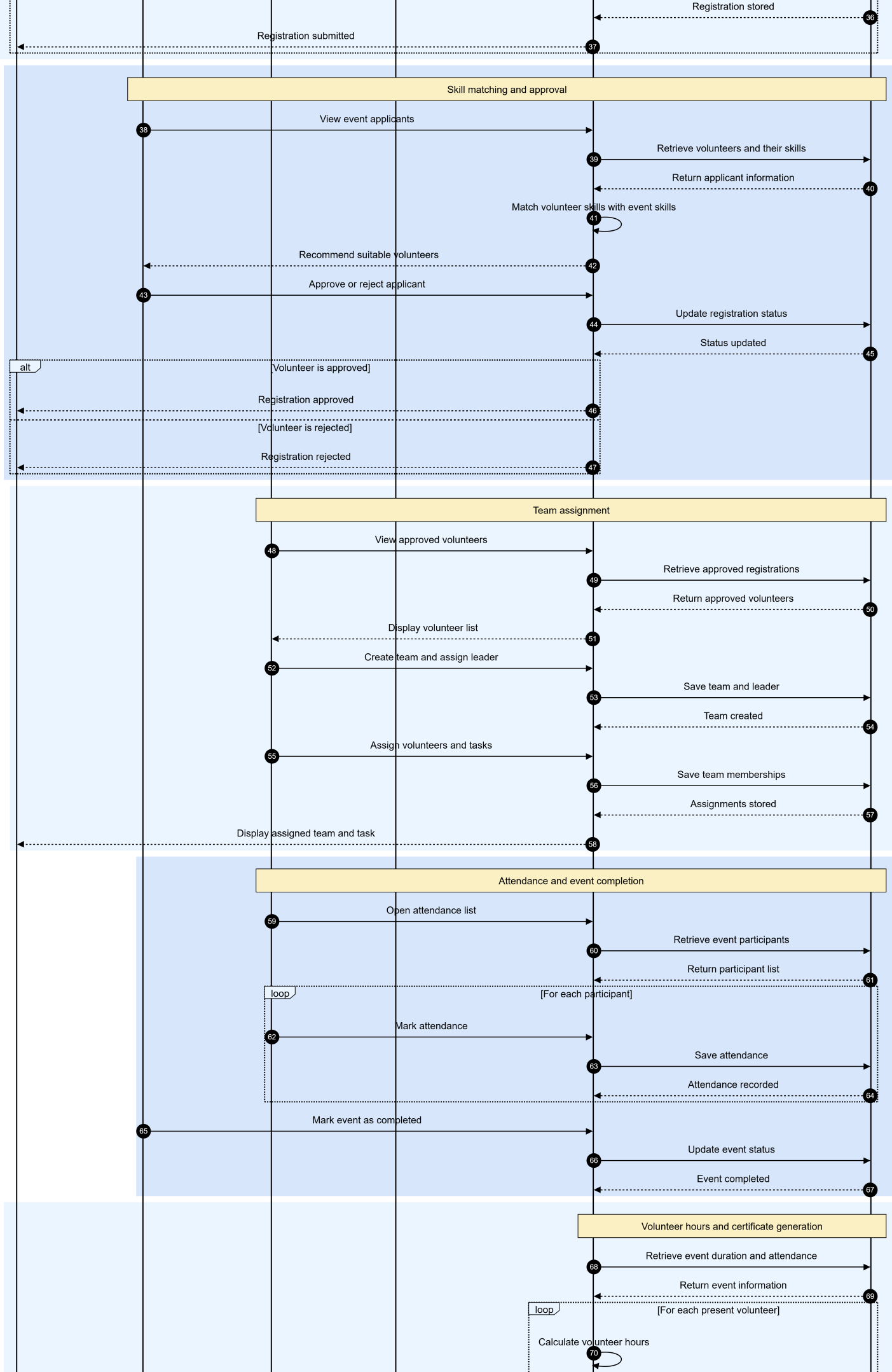
Github repo link :

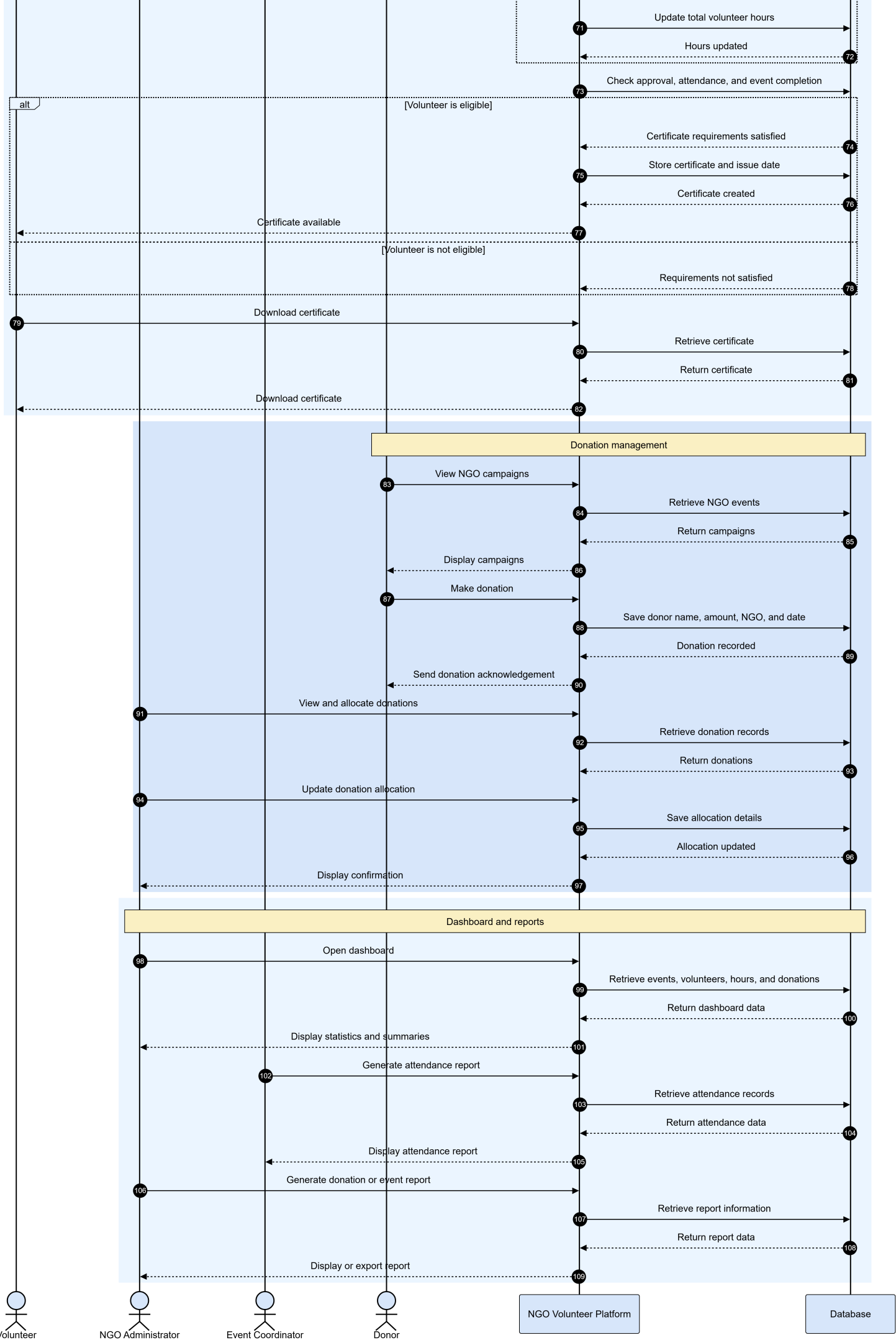
<https://github.com/FahimMuntasr/NGO-Volunteer-Coordination-Platform>











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